Marketing Analytics

Causal Comparisons and Policy Evaluation

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Session 10

A Repair-Policy Question

- A policy may change the cost of restoring restaurant availability.
- A national before-after change also includes common shocks.
- The target is the change attributable to the policy over a defined horizon.

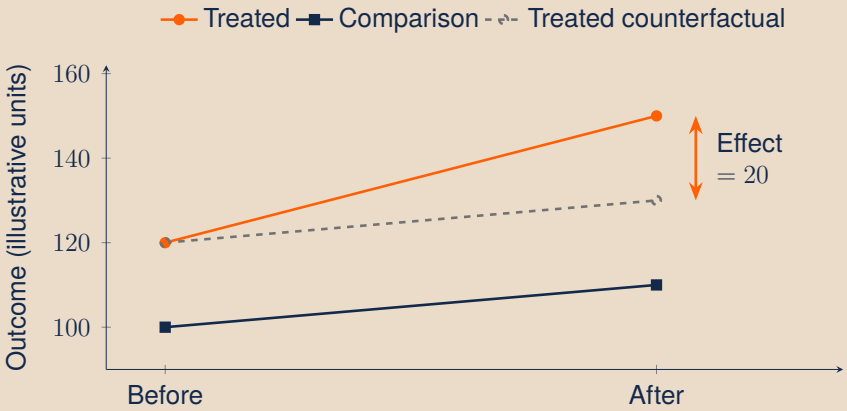
Difference-in-Differences

Subtract the comparison group's change from the treated group's change.

$$\widehat{\tau} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

Parallel untreated trends connect the observed changes to a causal effect.

The Counterfactual Trend

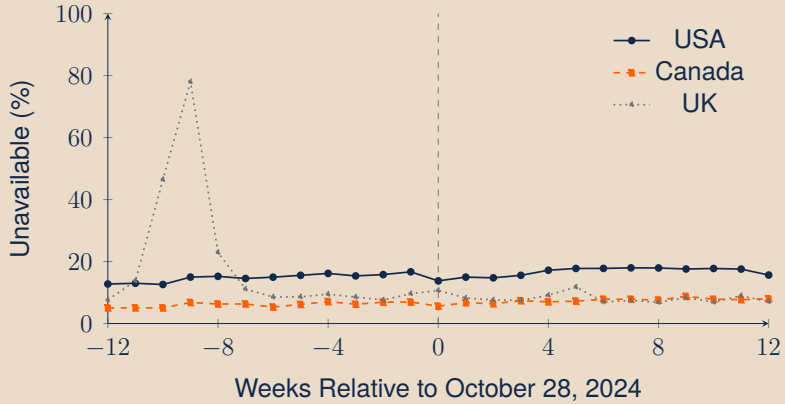


Synthetic outcomes. Treated sales rise 120 to 150 and comparison sales 100 to 110.

Design Checks

- Examine pre-treatment outcomes and institutional timing.
- Consider anticipation, spillovers, and changing composition.
- Match the comparison group to the policy's untreated counterfactual.

McBroken and the Comparison Group



A policy-analysis design requires a credible untreated comparison.

Regression Discontinuity

- Eligibility can change at a known score threshold.
- Nearby units require continuity of other outcome determinants.
- The resulting effect is local to the cutoff and compliance behavior.

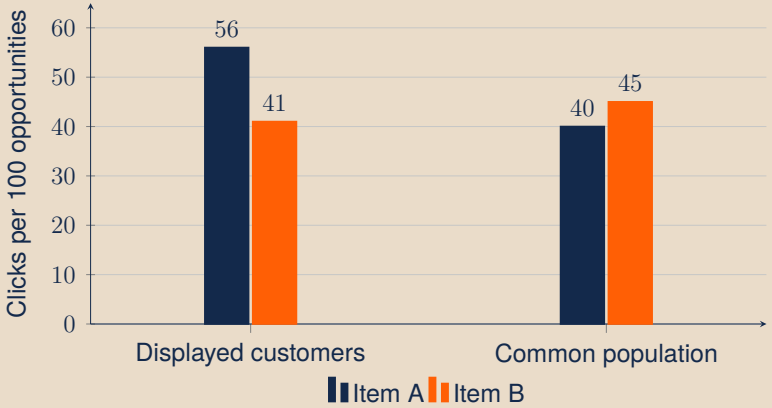
Instrumental Variables

- The instrument shifts treatment or price.
- Exclusion restricts its other paths to the outcome.
- Relevance alone does not establish exclusion.

Recommendation Logs Select Exposure

- Frequent buyers may usually see product A.
- Occasional buyers may usually see product B.
- Their raw click rates compare different customer populations.

Exposure Adjustment Reverses the Ranking



Synthetic randomized logging within frequent and occasional buyer groups.

Inverse-Propensity Policy Value

Weight each observed outcome by target-policy probability over logging probability.

$$\widehat{V}(\pi) = \frac{1}{n} \sum_i \frac{\pi(A_i \mid X_i)}{p_0(A_i \mid X_i)} Y_i$$

The denominator counts every eligible logged opportunity.

A Common-Population Calculation

Synthetic clicks and logging probabilities identify the two policy values.

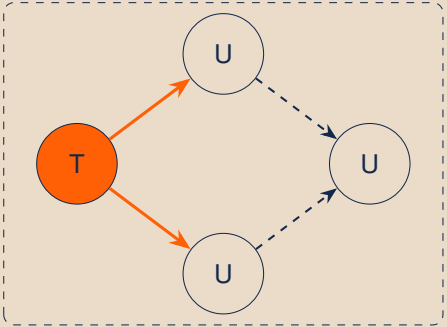
```
n = 1000
value_a = (270 / 0.90 + 10 / 0.10) / n
value_b = (25 / 0.10 + 180 / 0.90) / n
raw_a = 280 / 500
raw_b = 205 / 500
assert (value_a, value_b) == (0.40, 0.45)
```

The raw ranking and policy-value ranking disagree.

Support and Weight Concentration

- An action never shown to a group has no observed outcome there.
- Large weights amplify a small number of records.
- Clipping changes the estimator and its bias-variance tradeoff.

Spillovers Change the Outcome Comparison



One connected household or market can form a randomization cluster.

T: assigned treatment
U: assigned control

Control status does not guarantee absence of exposure through other people.

Synthetic store network with effects beyond assigned units.

Coding Exercise: Policy Value

- Reproduce raw rates and inverse-propensity values.
- Check them using within-group averages.
- Set one logging probability to zero and explain the missing comparison.

Information and Exposure

- A recommendation can draw attention to an item.
- A personalized endorsement can also change beliefs.
- Separate interventions can identify these channels.

A Policy Value for Whom?

- Clicks and sales can rise while customer value falls.
- A causal estimate concerns the outcome that was actually measured.
- Record refusal, completion, and customer time when procedures change.

Causal-Design Discussion

- What would the treated group have experienced without the policy?
- Which group or action lacks comparable support?
- What institutional evidence strengthens the proposed comparison?